



YISHUN INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
Higher 2

CANDIDATE
NAME

CG

DATE

H2 BIOLOGY

9744/04

31 Aug 2021

2 hours 30 minutes

READ THESE INSTRUCTIONS FIRST

Write your name and class in the spaces at the top of this page.

Write in dark blue or black pen only.

You may use a soft pencil for any diagrams or graphs.

Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

Shift
Laboratory

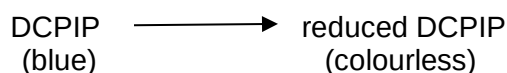
For Examiner's Use	
1	36
2	19
Total	55

This document consists of **20** printed pages.

Answer **all** questions.

QUESTION 1

DCPIP is a redox dye that can be used to study photosynthesis. It is reduced by accepting hydrogen, causing it to change from blue to colourless.



You are required to investigate the effect of light intensity on the rate of photosynthesis in spinach leaves using DCPIP.

You are provided with:

- spinach leaves
- cold buffer solution, labelled **C**
- DCPIP solution, labelled **D**

You are required to:

- decide the range of distances to vary light intensity investigated
- prepare leaf extract from spinach leaves
- test the effect of light intensity on rate of photosynthesis

A student conducted preliminary investigations, where light intensity was taken to be inversely related to the distance of the sample from the lamp, i.e. $\text{light intensity} = 1/\text{distance}$. In the investigation, the student varied the distances of the samples from the lamp and observed for DCPIP decolourisation within 10 minutes. The student found that:

- samples placed at distances below 30 mm from the lamp were exposed to high heat emitted from the lamp, resulting in denaturation of enzymes involved in photosynthesis,
- samples placed at distances beyond 150 mm from the lamp did not decolourise within 10 minutes.

(a) (i) State the distances that you will use in your investigation.

[1]

Proceed as follows.

- 1 Using the scalpel and white tile, cut out 4 pieces (approximately 3 cm x 3 cm) of spinach leaf. Avoid areas on the leaf where large veins are present. You may combine pieces from different leaves.
- 2 Chop the leaves into fine pieces and place them into the plastic specimen tube.
- 3 Using a syringe, add 3 cm³ of cold buffer solution **C** into the specimen tube.
- 4 Using the glass rod, carefully grind the mixture in the specimen tube to obtain a green leaf extract.
- 5 Place a Petri dish on the edge of the tile, as shown in Fig. 1.1, such that it is tilted.
- 6 Carefully decant the green leaf extract from the specimen tube into the Petri dish, using a glass rod to trap the leaf debris. Ensure minimal leaf debris enters the Petri dish.

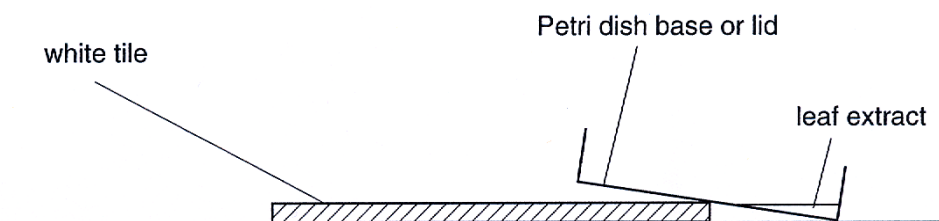


Fig. 1.1

- 7 Cover the Petri dish with aluminum foil loosely, but ensure its contents are completely shielded from light, except when removing samples. Keep the extract away from heat.
- 8 Place the lamp on the bench top so that it shines along the surface of the bench.
- 9 Place the A4-sized white paper next to the lamp such that the light source shines along the surface of the paper as shown in Fig. 1.2. You may elevate the paper to achieve this.



Fig. 1.2

- 10 On the paper, mark the distances from the lamp that you have selected in **(a)(i)**.

- 11** Fold the A4-sized black paper along its length to form a tent-like structure. Place the tent onto the white paper as shown in Fig. 1.3.

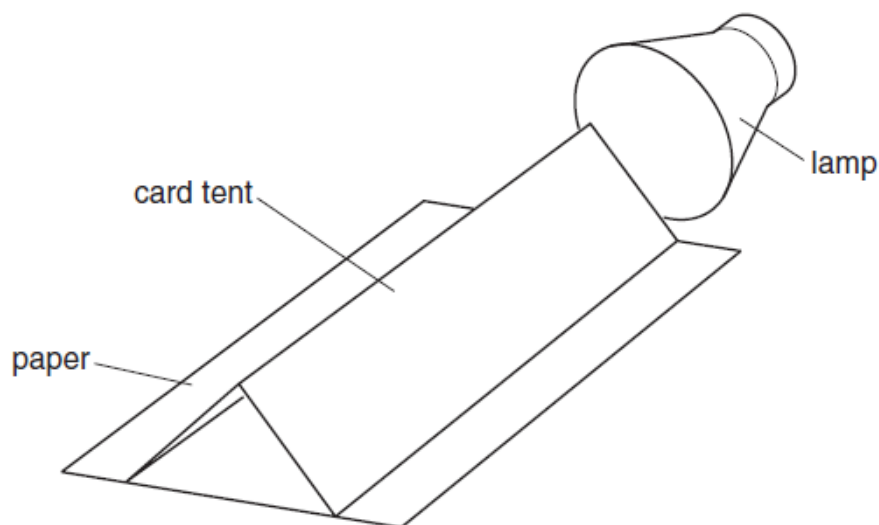


Fig. 1.3

- 12** Take a capillary tube and stand one end in the Petri dish to collect the leaf extract (about $\frac{3}{4}$ filled). Once filled, place the capillary tube containing the green leaf extract on the white tile and cover with aluminum foil. Label it tube **1**.
- 13** Using a teat pipette, add **2 to 3 drops** of DCPIP solution, **D**, to the leaf extract in the Petri dish. Swirl the Petri dish gently to mix the contents. Replace the foil cover.
- 14** Take the number of new capillary tubes for the number of distances that you have selected in **(a)(i)**.
- 15** Stand one end of each tube in the leaf extract/DCPIP mixture to fill $\frac{3}{4}$ of each tube. Cover the capillary tubes quickly with aluminum foil to minimise exposure to light.
- 16** Place one capillary tube at each of the distances marked on the white paper. Tape the capillary tubes down, and cover them with the card tent such that the only light source they are exposed to is the lamp.
- 17** Take one new capillary tubes and repeat **step 15** to fill the tube with leaf extract/DCPIP mixture. Label it tube **2** and leave the tube covered under aluminium foil.
- 18** Switch on the lamp and observe the colour of each of tubes on the white paper every minute for 10 minutes, comparing each tube to tube **1**. Measure the time taken for the DCPIP in each tube to completely decolourise. Record the time as '10' if the DCPIP does not decolourise.

- (ii) Use the space below to record your observations, in a suitable format.

The light intensity is given by the equation shown.

$$\text{light intensity} = \frac{1}{\text{distance from lamp}}$$

[4]

- (iii) The rate of photosynthesis is inversely proportional to the time taken for the DCPIP to completely decolourise.

Calculate the rate of photosynthesis at each light intensity investigated and add this data to your table in **(a)(ii)**.

Show your working for one light intensity in the space below.

[2]

- (iv) State the role of the following in the experiment:

Tube 1:

[1]

Tube 2:

[1]

- (v) Describe and explain another control that could be set up.

[2]

- (vi) Identify a potential risk in this experiment and the precaution you have taken.

[2]

- (vii) One significant source of error in this procedure is identifying when the colour in the capillary tube matches tube **1**.

Complete Table 1.1 to suggest:

- how to make an improvement in identifying when the colour in the capillary tube matches tube **1**
- one other significant source of error in this procedure
- how to make an improvement to reduce this other source of error.

Table 1.1

significant source of error	how to make an improvement
identifying when the colour in the capillary tube matches tube 1	

[3]

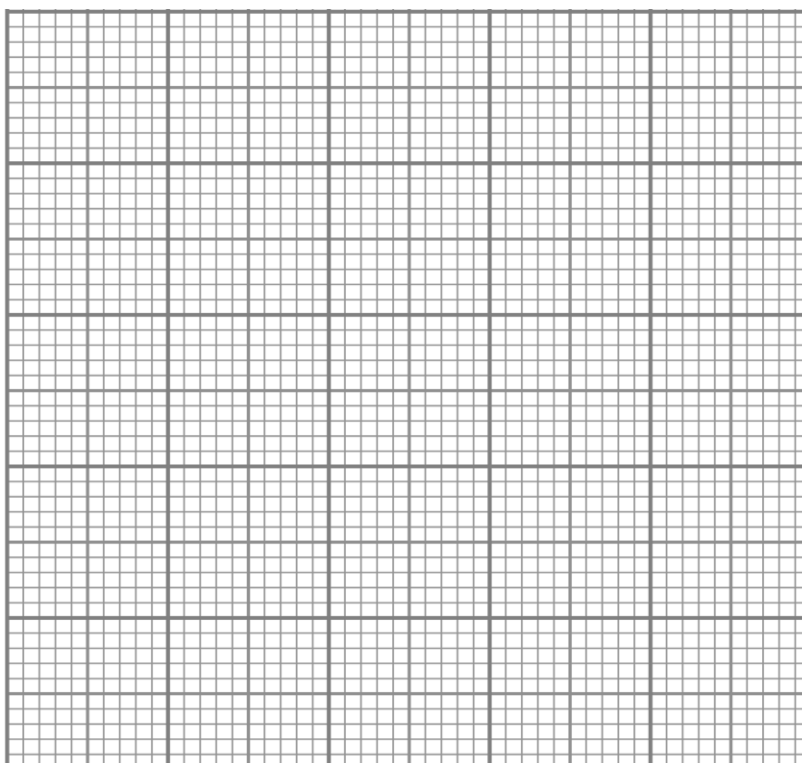
- (b) A student used a similar experiment to investigate the effect of wavelength on the rate of photosynthesis.

Table 1.2 shows the results of this investigation.

Table 1.2

wavelength / nm	rate of decolourisation of DCPIP / s ⁻¹
400	0.046
500	0.041
550	0.007
600	0.034
650	0.043

Plot a suitable graph to illustrate the relationship between the variables investigated.



[4]

- (i) With reference to the graph, describe and explain the relationship between wavelength and rate of decolourisation of DCPIP.

- (ii) Suggest how the different wavelengths can be obtained for the investigation.

In a separate experiment, a suspension of algal cells immobilised in alginate solution submerged in sodium hydrogencarbonate solution was used to determine the effect of carbon dioxide concentration on the rate of photosynthesis. Algal balls are unicellular algae that are immobilised in alginate.

As photosynthesis progresses, oxygen builds up in the air spaces in the algal balls over time.

Assuming that you have been provided with:

- a stock solution of 20.0 g dm^{-3} sodium hydrogencarbonate solution, **S** and
- distilled water, **W**

You are required to determine how to make a simple dilution of **S**, which reduces the concentration of **S** by 5.0 g dm^{-3} between each successive dilution. You are required to include 0.00 g dm^{-3} as one of your concentrations.

You will need to plan for the preparation of 20 cm^3 of each concentration.

(c) (i) Table 1.3 shows how to make up one of the concentrations.

Decide which other concentrations of **S** to prepare using simple (proportional) dilutions of **S** and complete Table 1.3.

Table 1.3

concentration of sodium hydrogen carbonate / g dm^{-3}	volume of S / cm^3	volume of distilled water, W / cm^3
20.0	20.0	0.0

[2]

(ii) Using the concentrations of **S** that you have determined in (i), plan an experiment to determine the effect of carbon dioxide concentration on the rate of photosynthesis.

You must use:

- algal balls, each with a diameter of approximately 5 mm

You may select from the following apparatus and use appropriate additional apparatus.

- normal laboratory glassware, e.g. test-tubes, boiling tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.
- syringes
- stopwatch
- bench lamp with 60W bulb.

[4]

- (d) Certain weed killers work by reducing the rate of photosynthesis. A student modified his experimental setup to investigate the effect of three different weed killers on algae at the same light intensity.

The investigation was replicated a number of times and the pH was recorded after 30 minutes. The student calculated the mean pH for each treatment, as shown in Table 1.4.

The student carried out *t*-tests to compare the mean pH for each of the treatments that used weed killers with the treatment that used no weed killer (treatment 1).

The student calculated the value of *t* for each of the tests.

The formula for *t*-test is:

$$t\text{-test} \quad t = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}} \quad v = n_1 + n_2 - 2$$

key to symbols

s = standard deviation

$\bar{x}$ = mean

n = sample size (number of observations)

v = degrees of freedom

Table 1.4 shows the results from their investigation.

Table 1.4

treatment	contents	mean pH after 30 minutes	<i>p</i> value
1	algal balls, indicator and distilled water	9.2	
2	algal balls, indicator and weed killer A	8.5	< 0.001
3	algal balls, indicator and weed killer B	9.0	> 0.05
4	algal balls, indicator and weed killer C	8.7	< 0.05

- (i) Suggest a null hypothesis for comparing the effect of treatment 2 with treatment 1.

.....

..... [1]

- (ii) The t -tests were used to compare the means.

State **one** feature of data that allows use of the t -test.

.....

..... [1]

- (iii) When calculating the effect of weed killer **A** on the rate of photosynthesis, 15 degrees of freedom was chosen. The students carried out 7 replicates of treatment 1.

State how many replicates were carried out for treatment **2**.

.....

..... [1]

- (iv) The students concluded that the weed killers had a significant effect on the rate of photosynthesis.

Explain to what extent the p values in Table 1.4 support their conclusion.

.....

.....

.....

.....

.....

.....

.....

.....

..... [3]

[Total: 36]

QUESTION 2

During this question you will require access to a microscope and access to slide **K1**.

K1 is a slide of a stained transverse section through a plant stem.

You are not expected to be familiar with this specimen.

(a) Observe all the different tissues in the stem on **K1**.

- (i)** Draw a large plan diagram of a quarter of the stem on **K1**, shown by the shaded area in Fig. 2.1.

Use a sharp pencil for drawing.

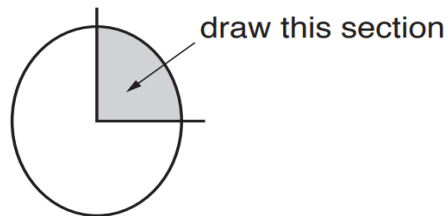


Fig. 2.1

You are expected to draw the correct shape and proportions of the different tissues.

- (ii) Observe the cells at the epidermis of the stem on **K1**.
The epidermis are cells which are at the edge of the stem.

Select **one** group of **four** adjacent, touching cells that make up this tissue.

Make a large drawing of this group of **four** cells.

Use **one** ruled label line and label to identify the cell wall of **one** cell.

You are expected to draw the correct shape and proportions of the different cells.

- (b) Fig. 2.2 shows a photomicrograph of a stained transverse section through part of the stem of a different plant species. You are not expected to be familiar with this specimen.

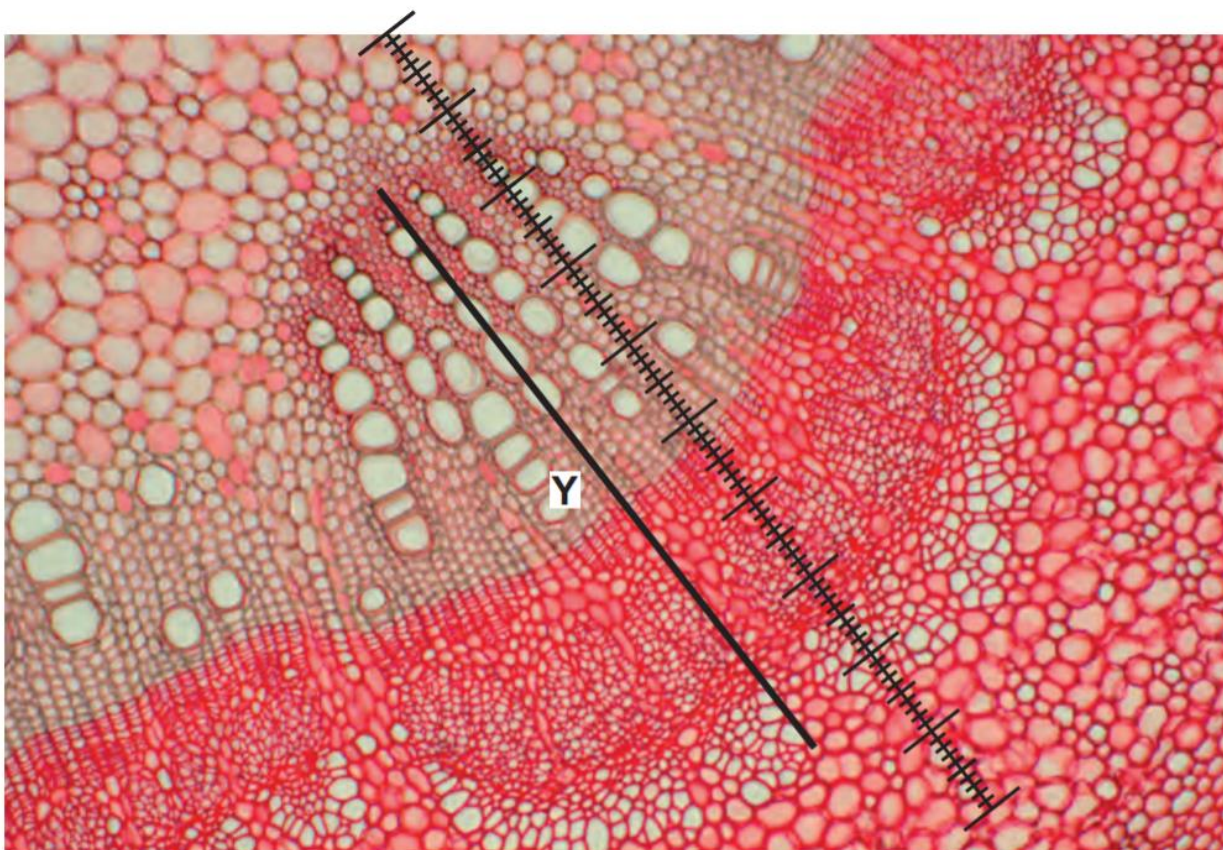


Fig. 2.2

A student calibrated the eyepiece graticule in a light microscope using a stage micrometer scale so that the actual length of the vascular bundle could be found.

The calibration was:
one eyepiece graticule division equal to 11 μm .

The photomicrograph in Fig. 2.2 was taken using the same microscope with the same lenses used by the student.

Use the calibration of the eyepiece graticule division and Fig. 2.2 to calculate the actual length of the vascular bundle, shown by line Y.

You may lose marks if you do not show all the steps in your working and do not use appropriate units.

Actual length of vascular bundle [3]

- (c) Some plant cells, for example in roots, store starch as grains.

Fig. 2.3 shows some different shapes of starch grains and patterns in the surface of the starch grains.

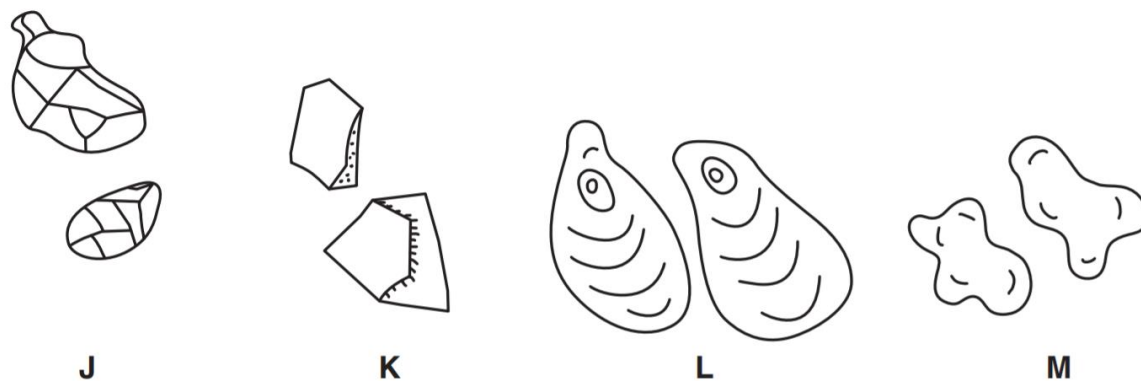


Fig. 2.3

You are required to:

- observe and draw the starch grains from potato cells
- observe the effect of iodine stain on starch grains.

You are provided with a piece of peeled potato tuber in a dish, labelled **T**, water in a container, labelled **W**, and iodine solution in a container labelled **I**.

Proceed as follows.

- 1 Put one clean and dry microscope slide on a piece of black card with a paper towel underneath as shown in Fig. 2.4.

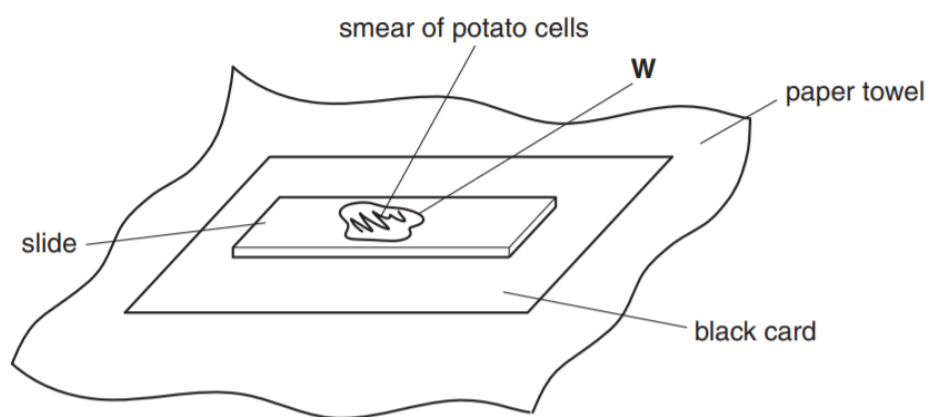


Fig. 2.4

- 2 Using a sharp blade or scalpel, cut the piece of potato to show a fresh surface.
- 3 Scrape the fresh surface of the piece of potato to remove a small quantity of tissue.

- 4 Put this tissue onto the slide and use the flat surface of the blade to squash this tissue.
- 5 Use forceps to remove any larger bits of tissue so that a smear of cells can be seen on the slide.
- 6 Put a few drops of **W** onto the smear of cells.
- 7 Cover the cells with a coverslip and use a paper towel to remove any excess liquid that is outside the coverslip.
- 8 View the slide to observe the starch grains using the microscope. Select an area of starch grains on the slide.

You may need to reduce the amount of light entering the microscope to observe cells clearly.

You must adjust the fine focus to observe the patterns in the starch grains.

You are required to use a sharp pencil for drawings.

- (i) Select **6** starch grains which show the different sizes and features of the grains.

Make a large drawing of these **6** starch grains.

[5]

- (ii) Use Fig. 2.3 to suggest which of the starch grains, **J**, **K**, **L** or **M**, matches some of the grains observed on the slide.

answer [1]

- 9 Remove the slide from the microscope and place it on a paper towel.

You will now add solution I to the water on the slide without removing the coverslip.

Put a drop of solution I onto the slide so that the drop touches the **edge** of the coverslip. Wait a few seconds while the solution I moves under the coverslip. Use the paper towel to remove any excess liquid from the top of the coverslip.

Immediately use the microscope to observe the starch grains where there is solution I on the slide.

- (iii) Suggest why using solution I as a stain may lead to inaccurate recording of the features of starch grains.

[1]

[Total: 19]